Deep vs. Classical Survival Models

Project Overview

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Main Question

Do deep survival models really beat classical ones once they are judged on more than ranking?

1 Summary of Each Literature Review

1.1 Literature Review 1: Nebiler

Calibration and Time-Ordered Validation of Deep Survival Models on Clinical Data

- **Main question:** how are deep survival models calibrated and checked on clinical data
- **Findings:** 7 of 9 studies use D-calibration; studies disagree on which measure shows calibration and on whether calibration costs ranking
- **Gap:** no study tests on later patients (all use random splits)
- **Niche:** train on earlier years, test on later years, compare Cox, random survival forest and a neural model on ranking and calibration

1.2 Literature Review 2: Sadraei

Competing Events and Calibration in Credit-Risk Survival Models

- **Main question:** how credit studies handle early repayment (a competing event) and check their probabilities
- **Findings:** 4 of 6 credit studies treat early repayment as a real outcome; 3 of 6 check probabilities, but none with a deep model; the only deep model (DeepHit) is judged on ranking alone
- **Gap:** survival-specific calibration tools such as D-calibration never reached the credit literature
- **Niche:** public mortgage data, earlier train and later test, compare classical models, random survival forest and DeepHit on ranking and probability

1.3 Literature Review 3: Becic

Survival Analysis: Benchmarking Machine and Deep Learning Models for Early Clinical Prediction

- **Main question:** what do benchmarks of survival models for early clinical prediction show
- **Findings:** papers proposing new deep models report superiority, but independent benchmarks often find random survival forest and Cox competitive or better; the choice of metric changes who looks best
- **Gap:** no common benchmark protocol; runtime is mostly omitted and tuning is poorly reported
- **Niche:** when do deep models win on ranking, calibration and runtime under a common evaluation protocol

1.4 What the Three Have in Common

- All compare deep models with classical ones (Cox, random survival forest); DeepHit appears in all three
- All separate ranking (C-index) from probability quality (Brier, calibration) and report that they can disagree
- All propose an experiment that compares classical and deep models on more than ranking
- Nebiler and Sadraei also share: same calibration definition, D-calibration and Brier as yardsticks, competing events as a gap, no or little testing on later data
- Becic adds: runtime, fair tuning, metric variants, a fixed rule for "best"

2 The Three Problems

One experiment with three switches: model type (Problem 1), split type (Problem 2), metric type (Problem 3). This keeps the workload manageable and shows which factor causes which change.

2.1 Focus: Competing Events

- The project focuses on competing events (e.g. death from another cause, early repayment of a loan) and leaves single-event survival out
- Reason: competing events was named as a gap in the literature reviews (Nebiler, Sadraei), and it is closer to real use
- Competing event: once one event happens, the others can no longer happen
- Treating the other event as censoring makes the event probability too high
- Consequence: DeepSurv drops out (single event only); the deep models come from DeepHit, Neural Fine-Gray, Deep Survival Machines or SurvTRACE
- Single-event survival is mentioned in the report as the simpler case covered by other work

2.2 Problem 1: Deep vs. Classical Models

- **Asks:** do deep competing-events models (e.g. DeepHit) beat cause-specific Cox, Fine-Gray and random survival forest when the comparison is fair
- **What each review adds:**
  - *Becic (main source):* results conflict between papers; new deep models are mostly tested by their own authors; tuning is hidden and runtime is missing. Adds to our design: the same tuning budget for all models, tuned on validation data only, with hyperparameters and seeds reported; runtime (training and prediction time) as an extra axis; the rule for "best" fixed before looking at results; established models only (cause-specific Cox, Fine-Gray, random survival forest, DeepHit)
  - *Nebiler:* deep models are mostly judged on ranking only; proposes comparing Cox, random survival forest and a neural model
  - *Sadraei:* the only deep model in credit studies (DeepHit) is judged on ranking alone, while classical and tree models give the probabilities

- **Test:** same data, same tuning budget for every model, several seeds, winner rule fixed in advance
- **Show:** main table of models by metrics; runtime vs. performance plot
- **Hypothesis:** deep models have a small ranking advantage that costs much more runtime

2.3 Problem 2: Testing on Later Data

- **Asks:** does the result of Problem 1 hold on later patients or loans
- **What each review adds:**
  - *Nebiler (main source):* none of the nine studies splits by time; random splits hide changes in patients and treatment over time
  - *Sadraei:* 3 of 6 credit studies test on later loans, so the idea is feasible in practice
  - *Becic:* no time-ordered testing. Adds to our design: its fairness rules (same tuning, reported seeds) apply to both split types, and results are reported by censoring level because rankings only matter within a censoring regime
- **Test:** same models twice, random split vs. time-ordered split (train early, validate next, test latest)
- **Show:** bar chart of the drop from random to time-ordered split per model
- **Hypothesis:** all models drop, deep models drop more

2.4 Problem 3: Ranking vs. Probabilities

- **Asks:** does a model that ranks well also give correct probabilities
- **What each review adds:**
  - *Nebiler (main source):* D-calibration type measures and Brier score as yardsticks; studies disagree on which measure shows calibration and on whether calibration costs ranking
  - *Sadraei:* D-calibration (Haider) exists but credit studies do not use it; the choice of ranking measure can change which model looks best
  - *Becic:* C-index and Brier score can disagree; standard vs. time-dependent C-index changes who looks best. Adds to our design: report both C-index variants next to the calibration measures
  - *All three:* DeepHit ranks well but has a poor Brier score
- **Test:** C-index (standard and time-dependent) next to D-calibration, integrated Brier score and calibration curves
- **Show:** ranking vs. calibration plot with one point per model; calibration curves at fixed horizons (e.g. 1, 3, 5 years)
- **Hypothesis:** the order of the models changes when calibration is added to ranking

3 Notes and Important Points

3.1 Planned Experiment

- Data: SEER breast cancer (death from breast cancer vs. other causes; has diagnosis years); mortgage data (e.g. Freddie Mac; default vs. early repayment) as optional second dataset, reported separately and not compared
- Models: cause-specific Cox, Fine-Gray, random survival forest for competing risks, DeepHit, plus one or two more deep competing-events models (Neural Fine-Gray, Deep Survival Machines, SurvTRACE)
- Metrics: standard and time-dependent C-index per event, Brier score on cumulative incidence, calibration curves of cumulative incidence, competing-events D-calibration (Alberge et al., 2026), runtime
- Splits: time-ordered (train early, validate next, test latest) plus random split as control
- Repeat with several seeds; bootstrap intervals on the test set
- Hypotheses:
  - H1: deep models keep their ranking advantage on later data
  - H2: deep models lose more on calibration
  - H3: any gain costs clearly more runtime
- Any outcome is a result: if classical models match deep ones, that also answers the question

3.2 Things to Check

- SEER needs a signed data use agreement, so check access early
- Problem 2 needs calendar dates; METABRIC, PBC and Framingham probably lack them, so check before using them
- Mortgage data: labels for default, early repayment and maturity must be built by us, and the data is very large (subsample)
- Competing-risks random survival forest is mostly available in R (randomForestSRC), so R may be needed next to Python
- Competing-events D-calibration is new (2026); check whether Alberge et al. released code, otherwise implement it

3.3 Scope and Risks

- Many parts: several models, two split types, runtime, optional second dataset
- Order of work: SEER first, mortgage data only if time allows
- Fallback: SEER only, mortgage data described as future work
- If both datasets are used, results are reported per dataset to see whether the pattern repeats, with no claims about domain differences

3.4 Limits to Mention in the Report

- Nebiler and Sadraei counts describe studies chosen for interest in calibration, not how often calibration is checked in the field
- Both reviews have narrow venue rules and title-only search (DBLP)
- Single-event survival is left out, so results do not cover that setting
- Only a few deep models are tested, so the results speak for those models and not for deep models in general